

**FULL MARKS: 20**

## ID:

[illegible]

2. Design a combinational circuit using comparator ICs and logic gates to determine the **smallest among three input numbers**:  $A$ ,  $B$ , and  $C$ . The circuit must satisfy the following specifications: 7 +  
1 + 2  
(CO5)  
(PO3)

- **Inputs:** Three binary numbers  $A$ ,  $B$ , and  $C$  (you are not required to consider the number of bits, assume they are compatible with the comparator IC).
- **Outputs:** Three output signals:
  - **A smallest:** High (1) if  $A < B$  and  $A < C$
  - **B smallest:** High (1) if  $B < A$  and  $B < C$
  - **C smallest:** High (1) if  $C < A$  and  $C < B$
- It is guaranteed that no two inputs are equal. Hence, exactly one of the outputs will be high at a time.
- **Comparator IC:** You may only use 2-input comparator blocks. Each comparator has two  $n$ -bit number inputs ( $X$  and  $Y$ ) and three outputs:  $X > Y$ ,  $X < Y$ , and  $X = Y$ , where  $X$  and  $Y$  are two  $n$ -bit numbers.  
You are not required to design the comparator IC, it may be represented as a block in your diagram.
- **Design Requirement:** Using these comparator IC blocks and any other component as you wish, draw the **block diagram** of a combinational circuit that produces the required outputs. You may assume ideal behavior and unlimited availability of logic gates and comparators.

**Note:** A block diagram means that you do not need to draw the internal structure of the comparator ICs. Instead, use labeled blocks showing inputs and outputs clearly.

Answer the following questions -

- a) Design the above-mentioned circuit.
- b) Explicitly mention the number of 2-number comparator ICs you have used.
- c) Briefly explain the working mechanism (avoid unnecessary details).
- d) Propose the most efficient solution in terms of the number of comparators used. [**Bonus 5 Marks**]



3. Given,

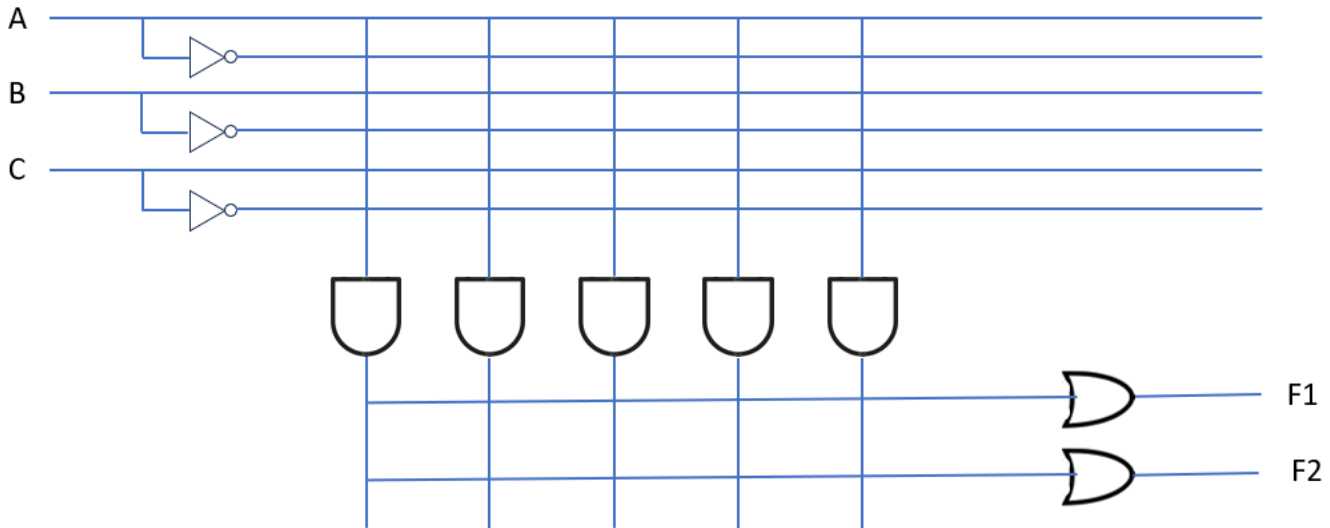
3.5 + 2

$$F_1(A, B, C) = \sum m(3, 5, 7)$$

(CO5)  
(PO3)

$$F_2(A, B, C) = \sum m(2, 3, 6)$$

Complete the PLD circuit for  $F_1$  and  $F_2$  by making the necessary connections (using 'X' at junctions) using the following **PLA (Programmable Logic Array)** structure.



**Figure 1:** PLA Circuit

And answer the following questions-

a) A PLA circuit structure is defined by  $n \times p \times m$ . Write down what each symbol represents and provide their values according to the given PLA circuit diagram that you have constructed.

- $n$ : \_\_\_\_\_, Value: \_\_\_\_\_
- $p$ : \_\_\_\_\_, Value: \_\_\_\_\_
- $m$ : \_\_\_\_\_, Value: \_\_\_\_\_

b) Put a tick next to the correct option.

For a **ROM (Read Only Memory)**:

- AND Array: ☐ Fixed ☐ Programmable
- OR Array: ☐ Fixed ☐ Programmable